

Hand-eye Calibration and Its Accuracy Analysis in Robotic Grinding

He Xie, Chang-tao Pang, Wen-long Li*, Yong-hua Li, Zhou-ping Yin

Abstract—Hand-eye calibration that establishes the pose relationship between a robot and a laser scanner is one of the most critical tasks in robotic grinding. Calibration accuracy can significantly affect the geometry and dimensional accuracy of parts such as blades. In this paper, a hand-eye calibration approach is proposed using a standard sphere as the calibration object. More important, three valid ways to improve the calibration accuracy are given. First, based on lots of experiment data, it's found that the scanning accuracy is related to the scanning area on the sphere. Combining error synthesis theory, an appropriate scanning area is found to improve the positioning accuracy of the sphere center. Second, we proved that moving along XYZ coordinate axes are best for robot during calibration process. Third, a novel orthogonalization algorithm of the calibrated rotation matrix is presented using differential kinematics. Different from the traditional orthogonalization algorithm, the algorithm converts the multiple solutions to a optimal approximate orthogonal solution, which avoids many unstable factors. Finally, test experiments show that the calibration accuracy is up to 0.0573mm. The calibration approach can be applied in robotic grinding of nuclear blades, aviation blades and turbine blades.

Keywords—hand-eye calibration; laser scanner; robotic grinding; orthogonalization

I. INTRODUCTION

With the development of science and industry, complex parts such as blades play a vital role in energy conversion products. Grinding is the last machining process to generate blade shape, which determines the manufacturing quality. Currently, grinding blades are mainly carried out by hand. Therefore, it is inefficient, characterized by high labor and environmental costs along with low efficiency. In contrast, robot methods are efficient and convenient, and could replace

This work was supported by the National Natural Science Foundation of China(No. 51475187), the National Basic Research Program of China(No. 2015CB057304), the Aeronautical Science Foundation of China(Grant No. 20140379001), and the Fundamental Research Funds for the Central Universities, HUST: 2015TS063..

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manual method for grinding blades [1-3]. In order to measure profiles in the robot end effector coordinate system rather than the scanner coordinate system during blade grinding, it is necessary to identify the pose relationship between the scanner and the robot. This identification is also called hand-eye calibration. Traditional calibration approaches calculated the rotation matrix and translation vector simultaneously by solving the homogeneous equation $AX=XB$ or $AX=YB$ [4-5]. Li et al. [6] proposed a hand-eye method to calibrate the pose relationship between a robot and a scanner. Yin et al. [7] enhanced the hand-eye calibration method by taking the robot positioning error into consideration. Staranowicz et al. [8] introduced a novel camera-calibration method for the estimation of the full set of intrinsic and extrinsic parameters. They all used known radius sphere as a standard object but did not analyze the positioning accuracy of the sphere center and didn't analyze the effect of robot's motion direction on calibration accuracy. Ren et al. [9] introduced a robot scanning system which expanded the scanning scope. However, the scanner was rigidly mounted on the robot end effector to calibrate the hand-in-eye pose.

This paper proposed a new hand-to-eye calibration approach in the application of robotic grinding. The rotation matrix and translation vector are calibrated separately using a standard sphere as a calibration object. The main contribution is that calibration accuracy is particularly analyzed by integrating theory with experiments. Main analyses are listed below.

- According to the accuracy experiment and error synthesis theory, the calibration accuracy is the highest when the linear scanning region is in the middle between the center and pole of the sphere.
- A novel orthogonalization algorithm of the rotation matrix is proposed using differential kinematics. Unlike the traditional orthogonalization algorithm, the proposed algorithm converts multiple solutions to a unique and optimal approximate orthogonal solution, which promotes the stability of calibration.

The structure of the paper is as follows. In the robotic belt grinding system, the reason to identify the pose relationship between the robot and laser scanner is given in Section II. A new hand-eye calibration approach is given in Section III and the calibration accuracy is analyzed in Section IV. The experiments are given in Section V. Conclusions are drawn in Section VI.

II. BACKGROUND

To know the significance of hand-eye calibration in robotic grinding, it's necessary to analyze the relationship between coordinate systems. In Fig.1, $\{S\}$, $\{B\}$, $\{E\}$, $\{T\}$ and

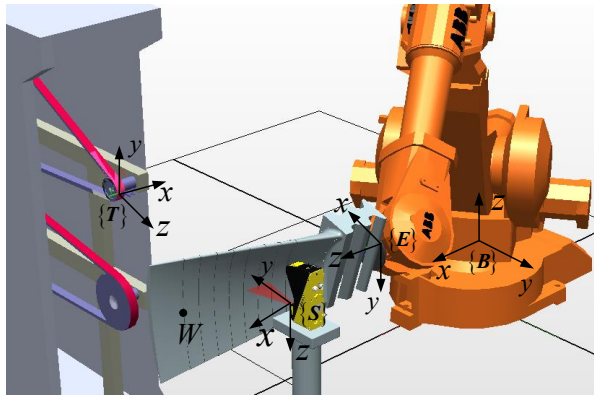


Fig. 1. Multi coordinates in the robotic belt grinding system

$\{W\}$, denote the laser scanner coordinate system (scanner system for short), robot basis coordinate system(basis system), end effector coordinate system(end effector system) , tool coordinate system(tool system) and workpiece coordinate system(workpiece system), respectively.

The origin of $\{T\}$ is on the contact wheel. The origin of $\{W\}$ is at the target point w that is on the blade CAD model. Both origins coincide so that the blade fit precisely with the belt. This coincidence requires calculating the transformation matrix wT that can be described by

$${}^wT = {}^wT {}^bT {}^eT {}^tT \quad (1)$$

where matrix bT represents a homogeneous transformation from the coordinate system α to β . Matrix tT can be obtained using linear voltage displacement transducer (LVDT) [10]. Matrix eT can be obtained from the robot controller. Matrix wT is unknown temporarily, but there exists transformation

$${}^wT {}^wP_i = {}^eT {}^bT {}^sP_i \quad (2)$$

where homogenous coordinates sP_i , wP_i represent the i -th measured point in the scanner system and the mapping point on the model, respectively. Matrix eT is known, and calibration of the hand-eye matrix bT will be introduced in the next section. According to (2), Matrix wT is determined by mapping between the model and measured points. Then, tT is determined using (1).

III. CALIBRATION

The hand-eye matrix bT establishes the pose relationship from the scanner system $\{S\}$ to the basis system $\{B\}$. This matrix bT is to be calibrated and can be decomposed as

$${}^bT = \begin{bmatrix} {}^bR & {}^bt_{so} \\ 0 & 1 \end{bmatrix} \quad (3)$$

where the rotation matrix bR represents the orientation from $\{S\}$ to $\{B\}$, and translation vector ${}^bt_{so}$ represents the distance from the origin of $\{S\}$ to $\{B\}$. Matrix bR and vector ${}^bt_{so}$ will be determined successively.

A. Position Measurement of Sphere Center

As a calibration object, a standard sphere is rigidly mounted on the robot end effector (Fig.2). For calibration, the sphere center is measured by circular fitting with the point set

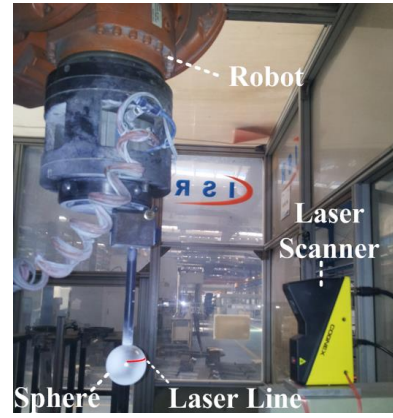


Fig. 2. Hand-eye calibration with a sphere

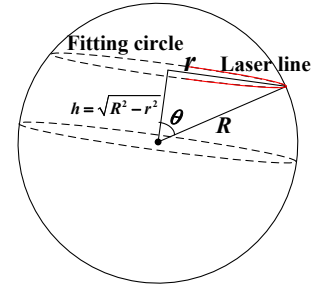


Fig. 3. Schematic measuring the sphere center

where the laser plane intersects with sphere (Fig. 3). Because the Z-coordinate values of points in the laser plane are zero, the position of the sphere center in $\{S\}$ is

$${}^sP = (x_s, y_s, \pm\sqrt{R^2 - r^2}) \quad (4)$$

where coordinate (x_s, y_s) represents the center of the fitted circle, symbol R and r represent the radiuses of the standard sphere and the fitted circle, respectively.

B. Orientation Calibration

Assume that the robot moves linearly along the given direction that is described by a unit vector bV in $\{B\}$. A set of reconstructed sphere centers ${}^sP = \{{}^sP_1, {}^sP_2, \dots, {}^sP_n\}$ in $\{S\}$ is obtained using (4). And a unit vector sV of the motion direction in $\{S\}$ is obtained by linear least squares fitting to sP . According to the transformation between linear spaces, a pose transformation from sV to bV can be described by

$${}^bV = {}^bR {}^sV \quad (5)$$

By moving the robot linearly along three different directions, the following equation is summarized from (5).

$$\begin{bmatrix} {}^bV_1 & {}^bV_2 & {}^bV_3 \end{bmatrix} = {}^bR \begin{bmatrix} {}^sV_1 & {}^sV_2 & {}^sV_3 \end{bmatrix} \quad (6)$$

The selection of three directions ${}^bV_1, {}^bV_2, {}^bV_3$ will be discussed in Section 4. (B). Define ${}^bV = [{}^bV_1 \ {}^bV_2 \ {}^bV_3]$ and ${}^sV = [{}^sV_1 \ {}^sV_2 \ {}^sV_3]$, and equation (6) can be rewritten as ${}^bV = {}^bR {}^sV$. Then the rotation matrix is calculated by

$${}^bR = ({}^bV)^{-1} {}^sV \quad (7)$$

For the scanning error and the motion error of the robot, bR in (7) is not orthogonal and should be orthogonalised. The traditional Gram Schmidt orthogonalization for bR has

multiple solution, but the solution is actually unique. And calculating the most approximate orthogonal matrix of ${}^b_s R$ will be introduced separately in Section IV. (C).

C. Positioning Calibration

With a sphere center as the feature point, establishing the coordinate relationship between ${}^s p$ in $\{S\}$ and ${}^e p$ in $\{E\}$ is needed to determine the translation vector ${}^b t_{so}$. According to the robot kinematics, there is a space transformation from $\{S\}$ to $\{E\}$ through $\{B\}$, and the coordinate relationship can be defined by

$$\begin{bmatrix} {}^e p \\ 1 \end{bmatrix} = \begin{bmatrix} {}^e_b R & {}^e t_{bo} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^b_s R & {}^b t_{so} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^s p \\ 1 \end{bmatrix} \quad (8)$$

where the rotation matrix ${}^e_b R$ and the translation vector ${}^e t_{bo}$ represent the orientation and position from $\{S\}$ to $\{E\}$, respectively. They both can be obtained from the robot controller. The rotation matrix ${}^b_s R$ has been determined in the last subsection and determining the translation vector ${}^b t_{so}$ is the task in this subsection.

By moving the robot to n different poses and keeping the sphere in the field of the scanner view, a set point of sphere centers ${}^s P = \{{}^s p_1, {}^s p_2, \dots, {}^s p_n; n \geq 3\}$ in the scanner system is calculated using (3). By substituting ${}^s P$ into (8), n similar coordinate transformations are obtained.

$$\begin{bmatrix} {}^e p \\ 1 \end{bmatrix} = \begin{bmatrix} {}^e_b R_i & {}^e t_{boi} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^b_s R & {}^b t_{so} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^s p_i \\ 1 \end{bmatrix} \quad (9)$$

where $i = 1, 2, \dots, n$. The position of sphere center ${}^e p$ stays constant, because the sphere is rigidly mounted on the end effector.

By expanding (9), it can be rewritten as

$${}^e p = {}^e_b R_i {}^b_s R {}^s p_i + {}^e t_{boi} + {}^e_b R_i {}^b t_{so} \quad (10)$$

For simplification, let $q_i = {}^e_b R_i {}^b_s R {}^s p_i + {}^e t_{boi}$ be the constant term. Then, equation (10) can be rewritten as

$${}^e p = q_i + {}^e_b R_i {}^b t_{so} \quad (11)$$

By subtracting the two adjacent equations in (11), the following n equations are obtained.

$$({}^e_b R_i - {}^e_b R_{i+1}) {}^b t_{so} = -(q_i - q_{i+1}) \quad (12)$$

Where $i = 1, 2, \dots, n-1$. Then translation vector ${}^b t_{so}$ can be solved by (12).

IV. ACCURACY ANALYSIS

Calibration accuracy is mainly affected by the calibration approach, scanning accuracy and the repeatability positioning accuracy of the robot. Here, three ways to improve the calibration accuracy are presented.

A. Positioning Accuracy of Sphere Center

The scanning accuracy ε_s is affected by the position of a

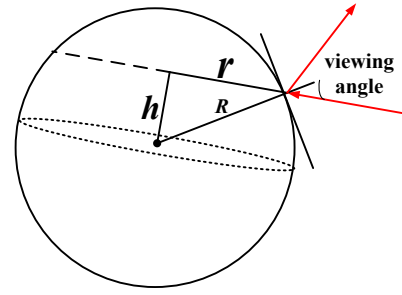


Fig. 4. Schematic of laser projection to sphere surface

scanning region on the sphere. The higher the height between the scanning region and sphere center is, the greater the viewing angle of the laser plane becomes and the wider the projected stripe becomes. The wider stripe can lower the scanning accuracy ε_s . It's concluded that scanning accuracy ε_s is relative to the height h in Fig(4) and the mathematical statement is $\varepsilon_s = \varepsilon_s(h)$.

The scanning accuracy ε_s can be defined as the radius standard deviation (SD) of the fitted circle. Assume that $P_a = \{p_{a1}, p_{a2}, \dots, p_{ai}, \dots, p_{am} \mid p_{ai} = (y_{ai}, x_{ai}); i \geq 3\}$ is the point set of the measured arc where the laser plane intersects with the sphere. ε_s can be calculated as

$$\varepsilon_s(h) = \sqrt{\frac{\sum_{i=1}^m [(x_s - x_{ai})^2 + (y_s - y_{ai})^2 - (R^2 - h^2)]}{m}} \quad (13)$$

Using (13), $\varepsilon_s(h)$ can be obtained by experiment. In the calibration method, all measured point sets P_a of arcs are used to calculate the sphere center coordinate ${}^s p$. Therefore, the position error of ${}^s p$ directly affects the accuracy of calibration. Because the 2D coordinate (x_s, y_s) of ${}^s p$ is also the coordinate of the fitted circle in the scanner system, XY coordinate errors $\Delta x_s, \Delta y_s$ of ${}^s p$ can be regard as the fitting accuracy or scanning accuracy ε_s approximately. Both errors can be described by

$$\begin{aligned} \Delta x_s &= \varepsilon_s(h) \\ \Delta y_s &= \varepsilon_s(h) \end{aligned} \quad (14)$$

According to the measuring error theory, the Z-direction error Δz_s of ${}^s p$ is defined by

$$\Delta z_s = \left\| \frac{\partial z_s}{\partial r} \right\| \Delta r = \frac{\sqrt{R^2 - h^2}}{h} \varepsilon_s(h) \quad (15)$$

where Δz_s , as with direction errors Δx and Δy , is a function of the height h . According to (4) and error synthesis theory, the position error $\|\Delta p\|$ of ${}^s p$ is defined by

$$\|\Delta p\| = \sqrt{\Delta x_s^2 + \Delta y_s^2 + \Delta z_s^2} \quad (16)$$

Equation (16) represents the fact that the combined error consists of XYZ direction errors $\Delta x_s, \Delta y_s$ and Δz_s . Reducing the position error $\|\Delta p\|$ can directly improve the calibration accuracy. Substituting (14) and (15) into (16), the minimizer $\|\Delta p\|$ can be solved by the equation.

$$\frac{\partial \|\Delta p\|}{\partial h} = 0 \quad (17)$$

B. Directions of the Linear Motion of a Robot

Assume that there are perturbations $\delta(^sR)$, $\delta(^sV)$ and $\delta(^bV)$ in the matrix sR , sV and bV , respectively. According to (7), the perturbation equation is defined by

$$\begin{bmatrix} ^bR + \delta(^bR) \end{bmatrix} \begin{bmatrix} ^sV + \delta(^sV) \end{bmatrix} = \begin{bmatrix} ^bV + \delta(^bV) \end{bmatrix} \quad (18)$$

According to perturbation theory and (18), the relative error of bR is

$$\frac{\| \delta(^bR^{-1}) \|}{\| ^bR^{-1} \|} \leq \frac{\text{cond}(^sV)}{1 - \text{cond}(^sV)^s\kappa} (^b\kappa + ^s\kappa) \quad (19)$$

where $^s\kappa = \| \delta(^sV) \| / \| ^sV \|$, $^b\kappa = \| \delta(^bV) \| / \| ^bV \|$ and $\| \delta(^bR^{-1}) \| / \| ^bR^{-1} \| \approx \| \delta(^bR) \| / \| ^bR \|$. $\text{cond}(^sV)$ is the condition number of sV and $\| \delta(^bR) \| / \| ^bR \|$ is the relative accuracy of the orientation calibration. Relative error $^s\kappa$ and $^b\kappa$ are determined respectively by the accuracy of sphere center and by the robot's position repeatability. Thus, the maximum of the relative error $\| \delta(^bR) \| / \| ^bR \|$ requires the minimization of $\text{cond}(^sV)$.

The condition number $\text{cond}(^sV) \geq 1$, because sV is nonsingular. And the condition number $\text{cond}(^sV)$ is minimized if sV is orthogonal. When the robot moves linearly along three directions of axes XYZ alternately, the matrix sV is approximately orthogonal. Meanwhile the relative error $\| \delta(^bR) \| / \| ^bR \|$ is minimized and the way is simple to operate.

C. Orthonormalization with Iterative Optimization

To calculate the optimal approximate orthogonal solution $^bR_*^\perp$ of bR , the implementation steps are as follows.

1) Transform bR to an approximate Euler angle vector E_0 .

a) Calculate a approximate orthogonal matrix $^bR_0^\perp$ of bR using Gram-Schmidt algorithm. $^bR_0^\perp$ can be represented as

$$^bR_0^\perp = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, \text{ respectively.}$$

b) Vector $E_0 = [\alpha_0 \ \beta_0 \ \gamma_0]^T$ is the orientation of XYZ axes from the basis system to the scanner system. According to the fundamental definition of Euler angles, E_0 is calculated as

$$\left. \begin{aligned} \beta_0 &= \text{Atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}) \\ \alpha_0 &= \text{Atan2}(-r_{21}, r_{11}) \\ \gamma_0 &= \text{Atan2}(-r_{32}, r_{33}) \end{aligned} \right\} \quad (20)$$

2) Calculate the differential error vector δ_0 between E_0 and E_* :

Assume that there is a differential error vector $\delta_0 = [\delta_x \ \delta_y \ \delta_z]^T$ between the most approximate Euler angle vector E_* and the initial vector E_0 . Then the

corresponding differential error matrix between bR and $^bR_0^\perp$ is

$$s(\delta_0) = \begin{bmatrix} 1 & -\delta_z & \delta_y \\ \delta_z & 1 & -\delta_x \\ -\delta_y & \delta_x & 1 \end{bmatrix}$$

Then, the relationship between $^bR_0^\perp$ and bR can be defined by

$$s(\delta_0) ^bR_0^\perp = ^bR \quad (21)$$

By extracting δ_0 to a new linear equation, (21) can be rewritten as

$$A\delta_0 = B \quad (22)$$

where $A_{9 \times 3} = \begin{bmatrix} \mathbf{0}_{3 \times 1} & r_3 & -r_2 \\ -r_3 & \mathbf{0}_{3 \times 1} & r_1 \\ r_2 & -r_1 & \mathbf{0}_{3 \times 1} \end{bmatrix}$, $r_i = [r_{i1} \ r_{i2} \ r_{i3}]^T$ and

$B = \text{Vec}\left(\left(^bR_0^\perp - ^bR\right)^T\right)$, $\text{Vec}(\square)$ is a vectorization operator. Then, the differential error δ_0 is calculated as

$$\delta_0 = (A^T A)^{-1} A^T B \quad (23)$$

Update the Euler angle vector by $E_1 = (E_0 + \delta_0)$.

3) Transform the Euler angle vector E_1 to an updated rotation matrix $^bR_1^\perp$.

4) Replace $^bR_0^\perp$ with $^bR_1^\perp$ and repeat steps 2-3 until δ_0 is smaller than the given threshold on the N -th time.

Finally, the most approximate orthogonal matrix is $^bR_*^\perp = ^bR_N^\perp$.

V. EXPERIMENT

The robot scanning system (Fig.2) consists of a ABB IRB6650s industrial robot and a COGNEX DS1000 linear laser scanner whose relevant parameters are shown as follows.

- Scanning resolution (Y): 0.007mm~0.040mm
- Scanning resolution (X): 0.07mm~0.15mm
- Field of view: 120mm~175mm

The scanner is fixed on the ground using thick pillar. A standard sphere of radius $R = 30\text{mm}$ is mounted on the robot end effector using magnetic base.

A. Positioning Accuracy of the Sphere Center

First, move the robot to keep the sphere in the view field of the scanner. Then move linearly along axis z in the basis system, and meanwhile measure 35 sets of arc points with proper space which is defined as the height h . Finally, a set point of 35 sphere centers sP is obtained using the method in Section III.A.

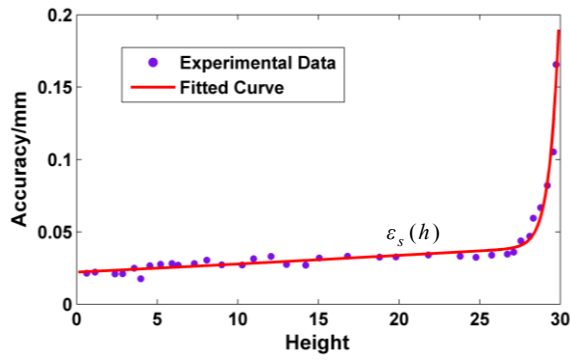


Fig. 5. Gaussian fitting between height h and fitting error ε_s

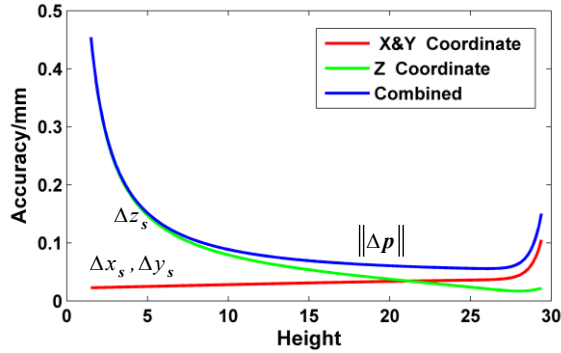


Fig. 6. Different accuracy with respect to height h

In Fig.5, the scanning accuracy ε_s with respect to height h is generated using (13) and sP , and the curve expression $\varepsilon_s(h)$ of Gaussian fitting is

$$\varepsilon_s(h) = 0.1718 \times 10^{12} \exp(-0.15726(h - 64.77)^2) + 0.0581 \exp(-0.01222(h - 80.28)^2)$$

The scanning accuracy ε_s is positive to the height h . The minimum of ε_s is 0.0227mm when the laser plane is nearly intersected with sphere center ($h \approx 0mm$). ε_s is acceptable when $0mm < h < 27mm$ (average $\bar{\varepsilon}_s = 0.031mm$), but unacceptable when $27 \leq h \leq 30$.

In fig.6, as with ε_s , errors $\Delta x_s, \Delta y_s$ are positively to height h . The error Δz_s is acceptable when $h \geq 19.2mm$ ($\Delta z_s < 0.04mm$). According to (17), the combined position error $\|\Delta p\|$, the most important error among them, is minimum when $h = 26.001mm$ and acceptable when $20mm \leq h \leq 28mm$; Thus, in order to improve the calibration accuracy, the height h between the sphere center and the laser plane should range from 20 to 28 as a whole.

B. Calibration

Guided by the scanning experiment results above, the robot is moved linearly along three axes XYZ, alternately. Then, bV is a unit matrix and three point sets sP are obtained. sV is calculated by liner fitting with sP . According to (7), rotation matrix bR is determined.

$${}^bR = \begin{bmatrix} 0.99951 & 0.00291 & 0.03132 \\ 0.02823 & -0.00707 & -0.99958 \\ -0.00268 & 0.99997 & -0.00716 \end{bmatrix}$$

For comparison, initial two initial Euler angle vectors are obtained using (20).

$$E_{10} = [0.028237 \quad 0.002709 \quad 1.577946]^T \&$$

$$E_{20} = [0.031342 \quad 0.002658 \quad 1.577956]^T$$

The iterative orthonormalization processes of the matrix $R_{10}^\perp$ and the matrix $R_{20}^\perp$ are recorded in Table I.

TABLE I. Iterative process of orthonormalization

	N	γ (rad)	β (rad)	α (rad)
Solution 1	0	1.577946	0.002709	0.028237
	1	1.577951	0.002684	0.029789
	2	1.577951	0.002684	0.029789
	3	1.577951	0.002684	0.029789
Solution 2	0	1.577956	0.002658	0.031342
	1	1.577950	0.002684	0.029789
	2	1.577951	0.002684	0.029789
	3	1.577951	0.002684	0.029789

Although, Euler angle vectors start to be different, but soon the same with iterative optimization. The most approximate Euler angle vector E^* and orthogonal matrix $R_\perp^*$ are shown as follows.

$$E^* = [1.577951 \quad 0.002648 \quad 0.029789]^T \&$$

$$R_\perp^* = \begin{bmatrix} 0.99955 & 0.00290 & 0.029765 \\ 0.02978 & -0.00707 & -0.99953 \\ -0.00268 & 0.99997 & -0.00715 \end{bmatrix}$$

By moving the robot to 32 large different poses, 32 pairs of data $({}^bR, {}^e t_{bo}, {}^s p)$ are obtained. Then translation vector is calculated using (10)-(12).

$${}^b t_{so} = [830.540 \quad 1609.471 \quad 1539.871]^T$$

C. Sphere Reconstruction

With the pose calibrated, reconstructing the sphere (Fig. 7) is feasible to test the position accuracy of calibration. The sphere center coordinate is ${}^e p = (-35.317, 21.6473, 413.6171)mm$ with standard deviation $e_{std} = 0.0573mm$. And the measured sphere radius is $R_{mea} = 29.987mm$ with 30mm standard radius. These results show good accuracy and stability).

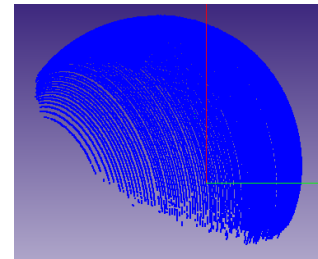


Fig. 7. Reconstructed sphere

VI. CONCLUSION

This article deals with the hand-eye calibration problem in the application of robotic grinding. In the proposed calibration approach, the rotation matrix and translation vector are calibrated separately using a standard sphere as a calibration object. In the accuracy analyses, a negative correlation between the scanning accuracy and the distance from the scanning region to the sphere center is found according to geometry optics theory. And the validity of the correlation is verified by the accuracy experiments. However, unlike the scanning accuracy, the calibration accuracy is first improved and then reduced along with the increase of the distance from the scanning region to the sphere center. Then, using proposed orthogonalization algorithm, rotation matrix is transformed iteratively from multiple solutions using Gram Schmidt algorithm to a unique and best solution. The proposed algorithm is fast and can enhance the calibration stability. Finally the scanning experimental results show that the position accuracy of calibration can reach to 0.0573mm, which meets the requirement of blade surface qualities. The calibration approach can be applied in robotic grinding of nuclear blades, aviation blades and turbine blades.

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